

Slow path (optimizer thread)

Cost model
(remaining order,
market state)

Execution QUBO
 $\min_z z^T Q z$

Solver: SA
(exact, QAOA offline);
uniform on failure

publish

Latest-value policy slot

state

Fast path (tick thread)

Market data
(bars / ticks)

Tick loop: poll slot
(never waits), re-plan
remaining shares

poll (non-blocking)

Child order
(TWAP fallback if
no policy yet)

